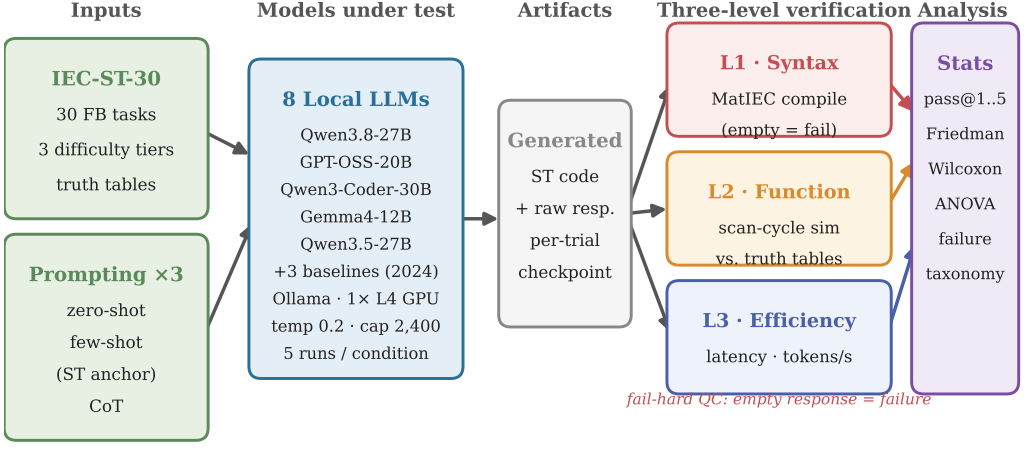
Inputs
Models under test
Artifacts
Three-level verification Analysis
IEC-ST-30
30 FB tasks
3 difficulty tiers
truth tables
Prompting ×3
zero-shot
few-shot
(ST anchor)
CoT
8 Local LLMs
Qwen3.8-27B
GPT-OSS-20B
Qwen3-Coder-30B
Gemma4-12B
Qwen3.5-27B
+3 baselines (2024)
Ollama · 1× L4 GPU
temp 0.2 · cap 2,400
5 runs / condition
Generated
ST code
+ raw resp.
per-trial
checkpoint
L1 · Syntax
MatIEC compile
(empty = fail)
L2 · Function
scan-cycle sim
vs. truth tables
L3 · Efficiency
latency · tokens/s
Stats
pass@1..5
Friedman
Wilcoxon
ANOVA
failure
taxonomy
fail-hard QC: empty response = failure